# CAR RENTAL MANAGEMENT SYSYEM

## SYSTEM DOCUMENTATION

**SUBMITTEDBY,**

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**DIT-01-0178/2025**

**A SYSTEM DOCUMENTATION SUBMITTED IN PARTIAL FULFILMENT FOR THE AWARD OF DIPLOMA IN INFORMATION TECHNOLOGY BY ZETECH UNIVERSITY**

**APRIL 2026**

***DECLARATION***

I hereby declare that this research project titled "Digital Transformation in Small Car Rental Businesses" is my original work and has not been submitted, either wholly or partially, for the award of any academic qualification in any other institution. Any materials, data, or ideas from other sources have been duly acknowledged and referenced in accordance with academic standards.

This project has been conducted under the guidance of the supervisor, and all ethical considerations pertaining to academic research have been observed.

Student Name: TRACY SOILA.

Registration Number: DIT-01-0178/2025.

Signature: _____

Date: _____

***DEDICATION***

This project is dedicated to my family for their unwavering support, encouragement, and patience throughout the development of this work. Their motivation and belief in my abilities have been a constant source of strength.

I also dedicate this project to my lecturers and classmates whose guidance, knowledge, and collaboration contributed greatly to the successful completion of this car rental management system.

***ABSTRACT***

this project focuses on designing and developing a digital solution aimed at improving the operations of small car rental businesses. Traditional car rental processes such as booking, record management, vehicle monitoring and customer communication are often manual time consuming and prone errors. The proposed system automates key processes including vehicle registration customer booking, payment tracking and report generation. A structured methodology was used, involving requirements analysis, system design, development, testing, and deployment. The system's feasibility was evaluated from technical, economic, operational, legal, social, and schedule perspectives. Findings show that the solution is viable and can significantly improve operational efficiency, reduce paperwork, enhance data accuracy, and support real-time decision-making. A Gantt chart was developed to guide project execution across the semester. The final system is expected to provide a more streamlined, reliable, and user-friendly platform that supports better service delivery in car rental businesses.

***DEFINITION OF KEY TERMS***

1. Digital transformation: the adoption and integration of digital technologies into business processes to enhance efficiency automate operations and improve customer experience in car rental services
2. Small car rental business: A rental enterprise with a limited fleet size, smaller staff, a localized operation, often serving a specific geographical area.
3. Online Booking System: A digital platform that enables customer to reserve vehicles remotely through a website or mobile application, reducing manual paperwork and walk -in processes.
4. Fleet management system: A software tool used to track, monitor, manage rental vehicles, including availability, maintenance schedules, fuel usage, and location.
5. Customer relationship management [CRM]: A digital system used to store client information, manage communication, follow up on bookings, and improve overall customer service.
6. Automation: The use of software tools and digital processes to perform repetitive tasks such as involving, and reservation updates without inputs.
7. Cloud computing: A technology that allows car rental businesses to store and access and improved data security

***ABBREVIATIONS AND ACRONYMS***

HTML–HyperTextMarkupLanguage

CSS – Cascading Style Sheets

ID – Identification.

PHP- hypertext preprocessor.

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## *CHAPTER ONE: PROJECT PLANNING AND ANALYSIS (WORKPLAN)*

### 1.1 Statement of Problem

According to microblinks (2025) small car rental businesses still face major challenges such as outdated processes, fragmented data and limited access to modern technologies. These practices result in operational inefficiencies, limited visibility of vehicle availability, and inconsistent service delivery, which negatively affect customer satisfaction and business performance. As the car rental market becomes increasingly digital, customers expect online booking, digital payments, and real-time information. Small car rental businesses struggle to meet these expectations due to limited resources, lack of technical expertise, and slow adoption of digital technologies. Industry researched by ispekt labs (2020) highlights that digital tools such as online booking, digital contracts, and automated inspections are transforming the car rental sector This creates a need for practical and affordable digital transformation solutions to enhance efficiency, competitiveness, and long-term sustainability.

### 1.2 Study Justification

This study is justified by the growing need for small car rental businesses to adapt to digital technologies in order to remain competitive in an increasingly technology-driven market. While large car rental companies have successfully adopted digital systems to improve efficiency and customer experience, small businesses often lag behind due to limited resources and technical challenges. Understanding how digital transformation can address operational inefficiencies is essential for improving service delivery and business performance in this sector.

Additionally, the findings of this study will provide practical insights into affordable and digital solutions suitable for small car rental businesses. The study will benefit business owners by

guiding informed decision-making, policymakers by highlighting areas for support, and researchers by contributing to existing literature on digital transformation in small enterprises. Ultimately, the study supports the sustainable growth and competitiveness of small car rental businesses.

## 1.3 System Objectives

### 1.3.1General Objective

To implement a digital car rental management system that improves operational efficiency, booking processes, and customer service in a small car rental business.

### 1.3.2 Specific Objectives

*1.* To develop an online Booking System for the customers to view available cars and make reservations digitally.

*2.* To build a digital Payment System that supports mobile money and card payments to reduce cash handling.

*3.* To implement a Customer Records Management system that store customer details, bookings, and rental history electronically.

*4.* To form a system that will record the condition of each car both at pick up and at return.

## **1.4 Functional Requirements**

| Users | Activities | System Features |
|---|---|---|
| **Staffs** | - Confirm and process bookings<br>- Check vehicles in and out<br>- Update vehicle status and maintenance<br>- Assist customers with inquiries<br>- Add, update, or remove vehicles<br>- Manage all bookings<br>- Generate reports (revenue, fleet usage)- Manage staff and user access | - Booking & Reservation System<br>- Vehicle Status Update<br>- Notifications (email/SMS)<br>- Customer Support Interface<br>- Vehicle/Fleet Management<br>-Booking Management<br>- Reporting & Analytic<br>- User Management |
| **Customer / Client** | - Browse available cars<br>- Make or cancel bookings<br>- Make online payments<br>- View booking history and status | - Online Booking & Reservation<br>- Payment Integration<br>- Booking History & Status<br>- Notifications & Reminders |

*Table 1.4 Functional Requirements.*

1.4.1Breakdown of Tools & Resources to Be Used

1. **Programming Languages:** PHP for backend HTML,CSS for frontend sdevelopment.
2. **Code Editor / IDE:** VS Code.
3. **Payment Gateway (Optional):** PayPal, Mpesa.
4. **Laptop:** Minimum 8GB RAM, Intel i5 processor, 256GB SSD, for development and testing.
5. **Internet Connection:** Stable broadband for research, deployment, and testing.
6. **Smartphone / Tablet:** For testing mobile responsiveness.
7. **Printer (Optional):** For invoices and reports.
8. **Developer (Student):** Responsible for system design, coding, and implementation.
9. **Car Rental Staff:** Provide requirements, test the system, and give feedback.
10. **Supervisor / Mentor:** Guide and evaluate the project.
11. **End-Users / Customers:** Test usability and practical functionality.

1.5 Project Schedule Breakdown

| | **PROJECT MILESTONES** | | | | | |
|---|---|---|---|---|---|---|
| **WEEKS** | **Project Planning & Analysis (System Documentation: Coverpage & Chapter One)** | **Project Design & Modeling (System Documentation Chapter Two)** | **Project Development & Testing (System Documentation Chapter Three)** | **Project Deployment (System Documentation Chapter four)** | **Final Touches of System Documentation (Preliminary Pages, Chapter Four & References)** | **Project Presentation** |
| 17-19 Jan | | | | | | |
| 20-26 Jan | | | | | | |
| 2-6 feb | | | | | | |
| 9-13 feb | | | | | | |

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | |
| 16-20 feb | | | | | | |
| 23-27 feb | | | | | | |
| 2-6 march | | | | | | |
| 9-13 march | | | | | | |
| 16-20 march | | | | | | |
| 23-27 march | | | | | | |

*Table 2project schedule breakdown*

Blue initially means that milestone has not achieved. Once you achieve that milestone, fill it with any other color of your choice to mean you're done with that part. That helps you to monitor your own progress versus the time scheduled.

When printing, this page must be printed in LANDSCAPE as the rest is printed in portrait

## *CHAPTER TWO: DESIGN AND MODELING*

### 2.1 Introduction to Modelling

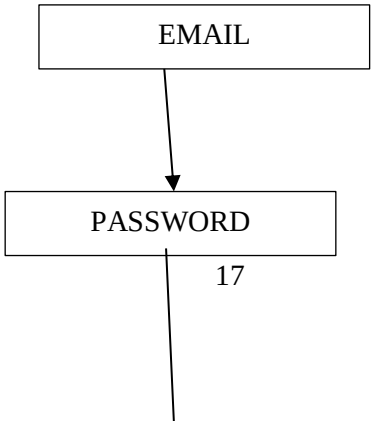
This chapter presents the designs and models used to visualize the car rental management system before development. Modelling was carried out to understand the system requirements, represent business processes, and design the structure of the proposed system. By creating models and diagrams in advance, it was possible to identify system components, user interactions, and data requirements early in the project. This approach reduced development errors and provided a clear blueprint for implementing the digital transformation of the small car rental business.

### 2.2 User Interface Models

This section presents the user interface (UI) models designed for the car rental management system. Before system development, all interfaces were hand-sketched and later refined using Figma to visualize the layout, workflow, and user interactions. These UI models represent the major system pages, forms, and interaction panels that support daily rental operations, customer management, and vehicle tracking.

#### 2.2.1 Login Page Design

The Login Page allows system users such as administrators and staff to securely access the car rental management system using their credentials. The page includes fields for username/email and password, as well as options for password recovery.



### 2.2.2 Sign Up Form Design

The Sign Up Form enables small car rental business owners or staff to create system accounts. The form collects essential details such as full name, email address, phone number, and password. Input validation is incorporated to ensure data accuracy.

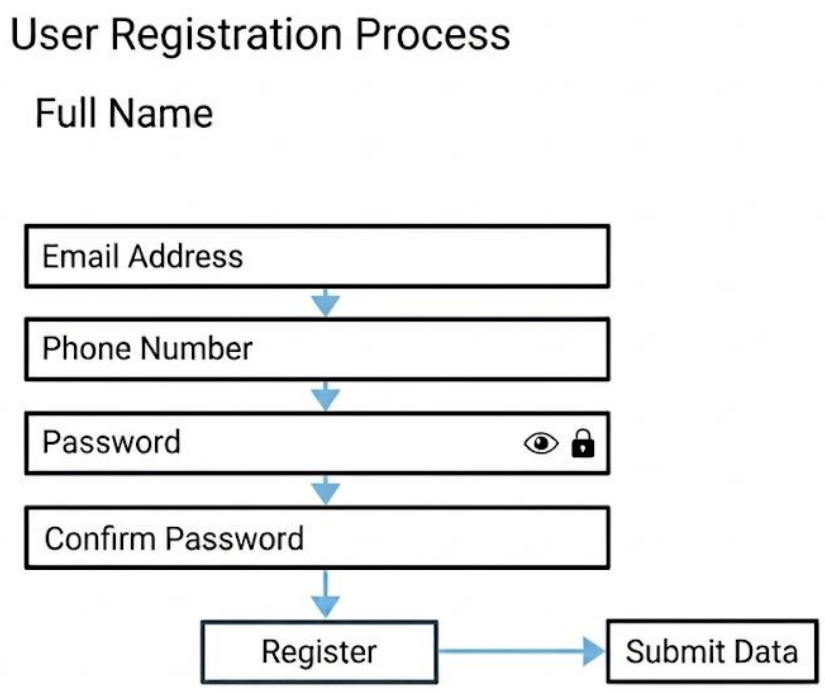



*Figure 2 sign up page design*

### 2.2.3 Admin Dashboard Design

The Admin Dashboard provides an overview of rental operations, including available vehicles, active rentals, pending returns, and revenue summaries. It serves as the central control panel for managing the car rental business efficiently.

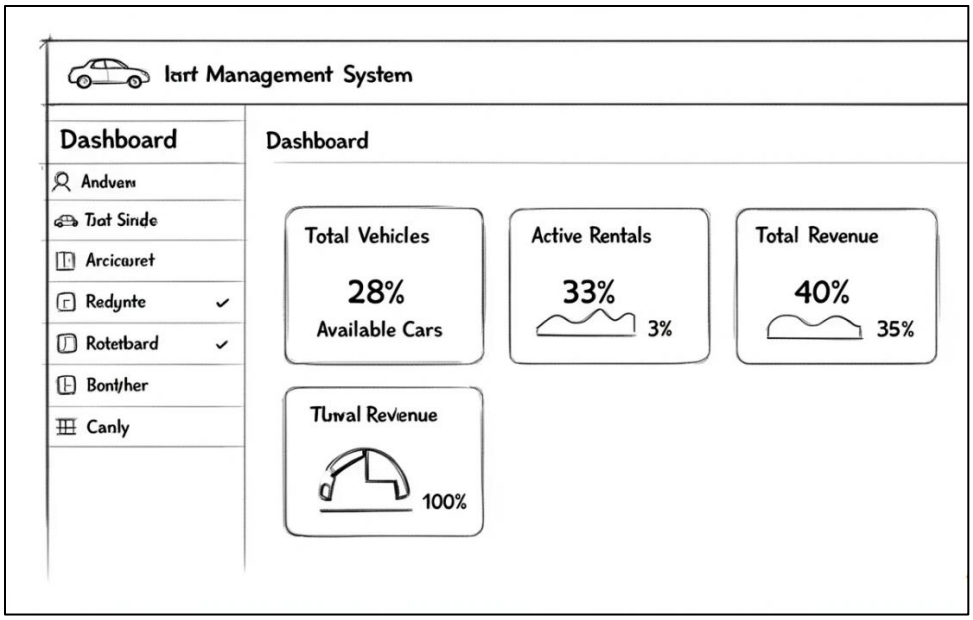



*Figure 3 Admin Dashboard Design*

2.2.4 Vehicle Management Page Design

The Vehicle Management Page allows administrators to add, update, and remove vehicle records. Information such as vehicle model, registration number, rental rate, and availability status is displayed in a structured format

| **Car Model** | **Reg Number** | **Price** | **Actions** |
|---|---|---|---|
| (Mazda) | XYZ 123 | $40/day | Edit / Delete |
| Toyota Camry | ABC 456 | Available | Available [Delete] |
| Honda Civic | ABC 456 | Rented | [Edit] / [Delete] |

| **Car Model** | **Reg Number** | **Price** | **Actions** |
|---|---|---|---|
| Ford F-150 | DEF 789 | Available | [Edit] / [Delete] |

***Vehicle Management Page Design***

2.2.5 Customer Management Page Design

This page is used to manage customer information, including personal details, rental history, and identification records. It helps the business maintain organized digital records instead of manual paperwork.

| **Customer Name** | **Phone Number** | **ID Number** | **Rental History** |
|---|---|---|---|
| (Empty) | (Empty) | (Empty) | [Delete] |
| **John Smith** | 555-123-4567 | DL12345 | Available |
| **Jane Doe** | 555-987-4563 | Rented | 3 Rentals (View) [Delete] |
| **Jane Doe** | PS98865 | 1 Rental (View) | Available |

*Table 2.5* ***Customer Management Page Design***

### 2.2.6 Rental Booking Page Design

The Rental Booking Page enables staff to create and manage car rental bookings. It allows selection of customers, vehicles, rental duration, and payment details, supporting a faster and more reliable booking process.

| **Field** | **Input / Selection** |
|---|---|
| **Customer Selection** | [ Select Customer] |
| **Vehicle Selection** | [ Select Vehicle] |
| **Start Date** | [ MM / DD / YYYY ] |
| **End Date** | [ MM / DD / YYYY ] |
| **Estimated Cost** | **$120** |
| **Action** | **[ CONFIRM BOOKING ]** |

Table 2.6 ***Rental Booking Page Design***

### 2.2.7 Payment and Invoice Page Design

This page displays payment details and automatically generates invoices for completed rentals. It supports digital payment tracking and improves financial transparency for small car rental businesses.

| Feature | Details / Input | Status / Action |
|---|---|---|
| **Customer Information** | John Smith (555-123-4567) | **Verified** |
| **Invoice Details** | Invoice #1024 | Date: 2026-03-02 |
| **Rental Item** | Toyota Camry (5 Days) | $40/day |
| **Financial Tracking** | **Digital Payment: Paid** | **Total: $200** |
| **Transparency Tools** | [ Generate Invoice ] | [ Download Receipt ] |

*Table 2.7* ***Payment and Invoice Page Design***

2.2.8 Reports and Analytics Page Design

The Reports and Analytics Page provides summarized insights such as total rentals, revenue trends, and vehicle utilization. These reports assist business owners in making data-driven decisions

| Metric Category | Key Insight / Data Trend | Business Impact |
|---|---|---|
| **Total Rentals** | 124 Bookings (Current Month) | Tracks business volume and growth. |
| **Revenue Trends** | **$4,960 Total** (+12% from last month) | Monitors financial health and cash flow. |
| **Vehicle Utilization** | 85% Fleet Active | Identifies which cars are making money. |
| **Top Performer** | Toyota Camry (Highest Demand) | Informs future fleet purchase decisions. |
| **Revenue Tiles** | $180 Avg. Transaction | Staff-only view for quick daily audits. |

*Table 2.8**8 Reports and Analytics Page Design***

## 2.3 Logic Models

This section presents logic models that visualize how the digital car rental system works. Diagrams were sketched to show system processes, data flows, and interactions before development.

### 2.3.1 User Authentication Flowchart

Shows the steps of user login and access verification.

CAR RENTAL MANAGEMENT SYSTEM-FLOW CHART

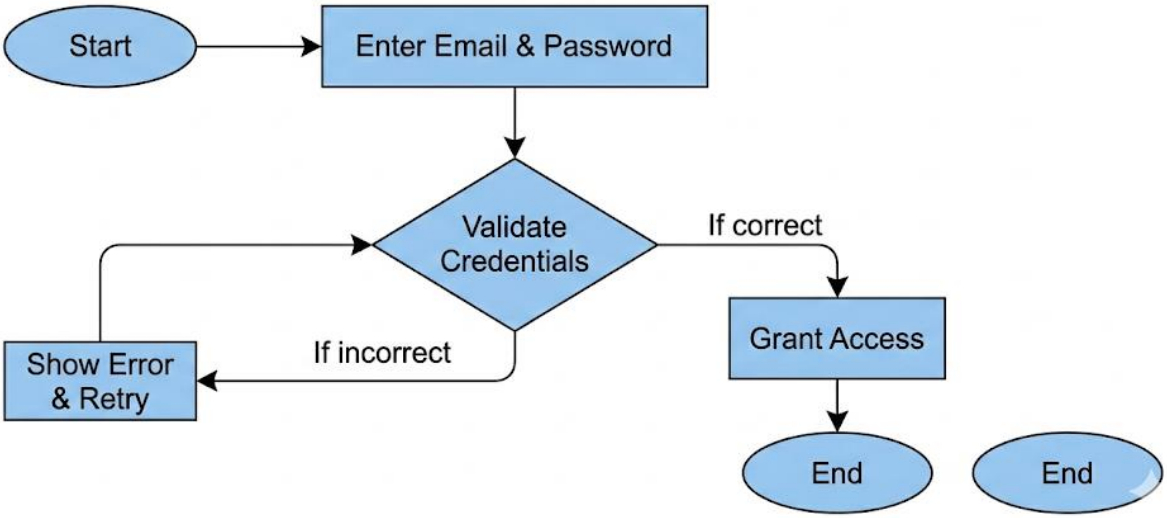



Table 2.9 ***User Authentication Flowchart***

### 2.3.2 Vehicle Rental Process Flowchart

Illustrates the vehicle rental process from selection to payment confirmation.

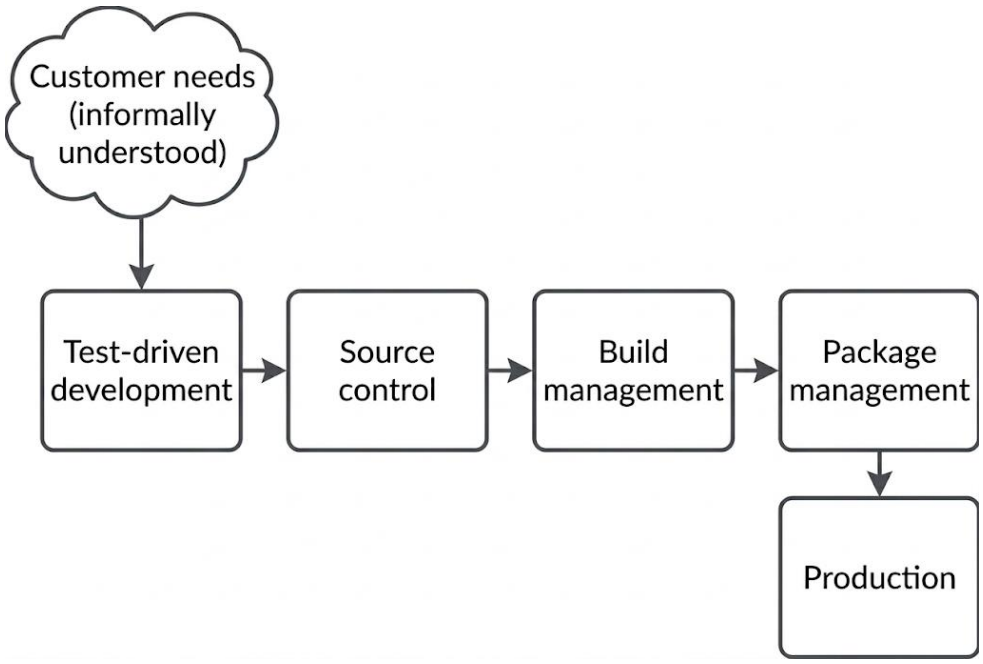



*Figure 4* ***Vehicle Rental Process Flowchart***

*2.3.3 Data Flow Diagram (DFD)*

Shows data movement between users, system processes, and the database.

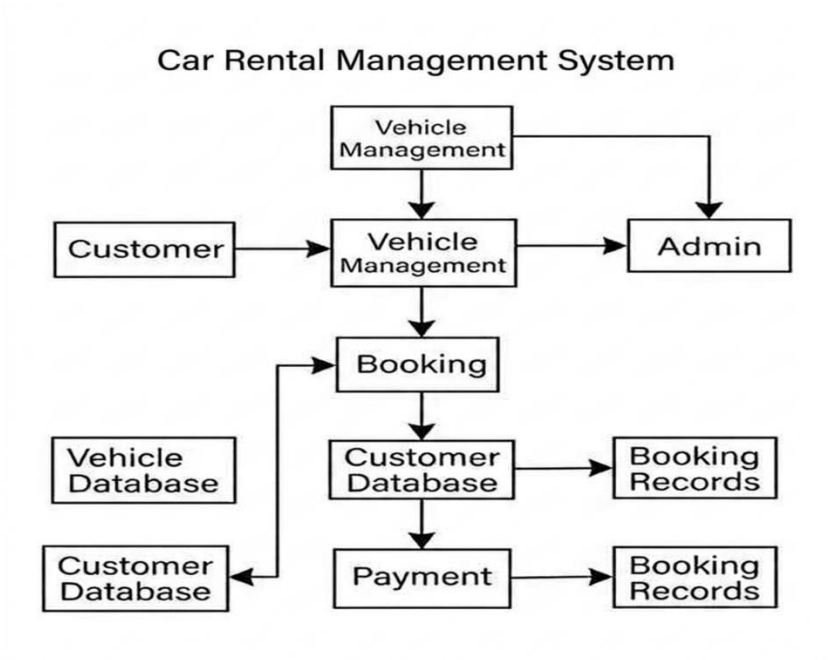



Table 3.1 ***Data Flow Diagram (DFD)***

2.3.4 Class Diagram

Depicts system classes (User, Vehicle, Customer, Booking, Payment) and their relationships.

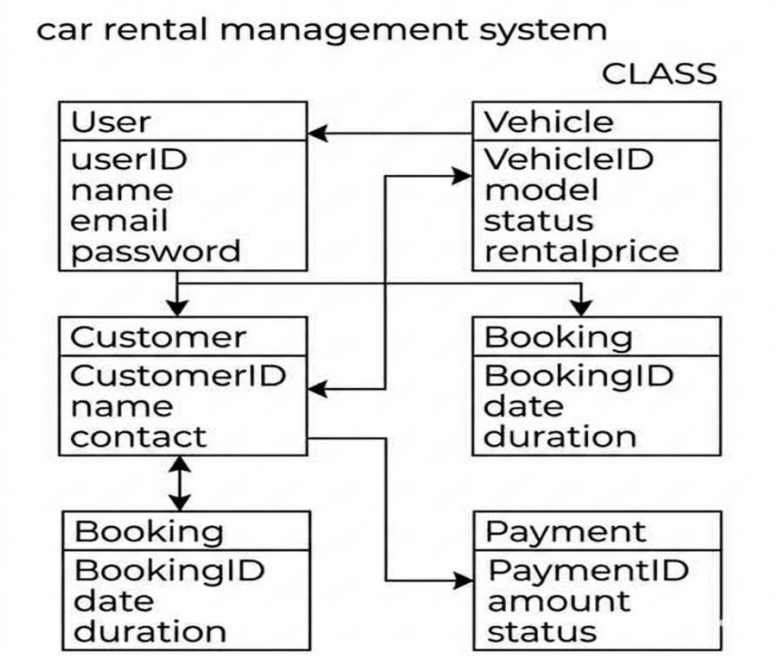



Table 3.4 ***Class Diagram***

2.3.5 Sequence Diagram for Booking

Demonstrates interactions between user, system, and database during a booking

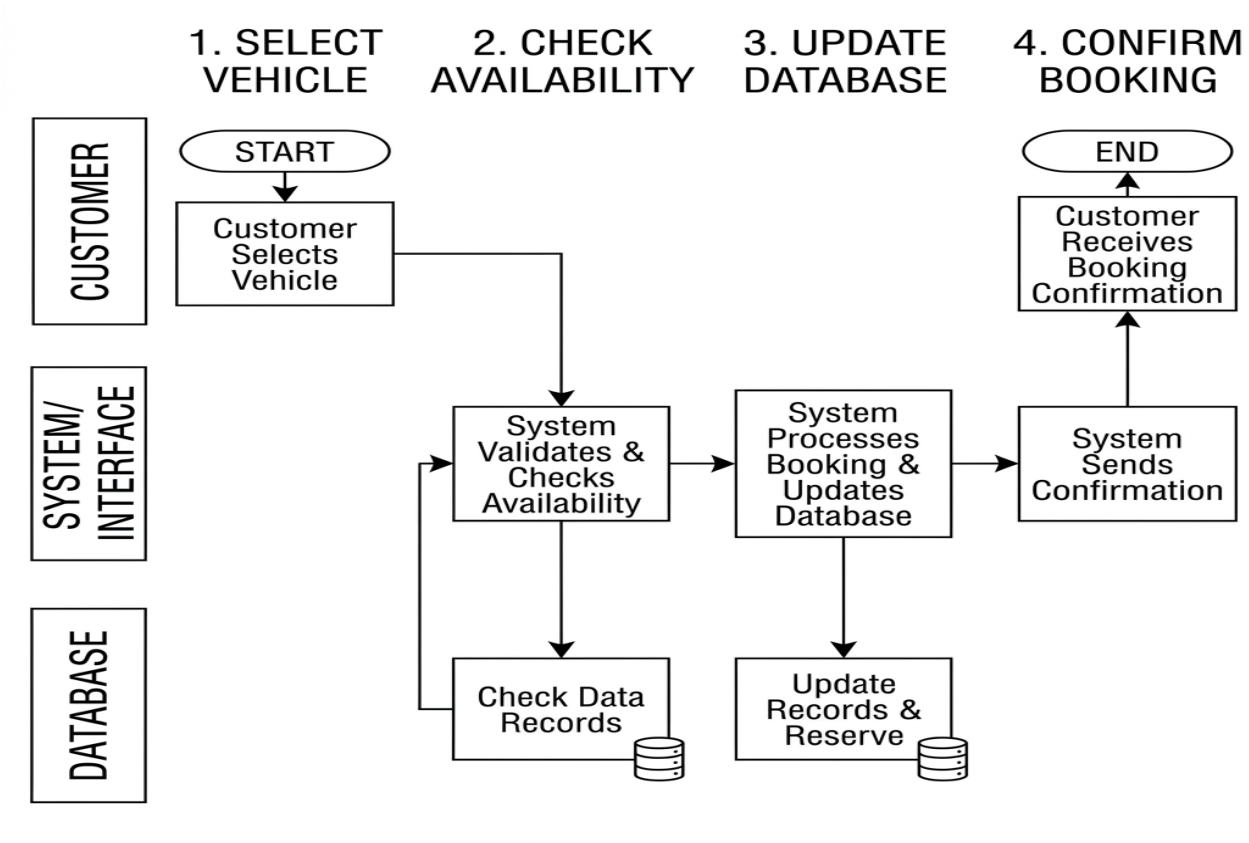



*Table 3.5* ***Sequence Diagram for Booking***

2.3.6 Vehicle Availability Decision Tree

Shows how the system checks vehicle availability based on dates and booking status.

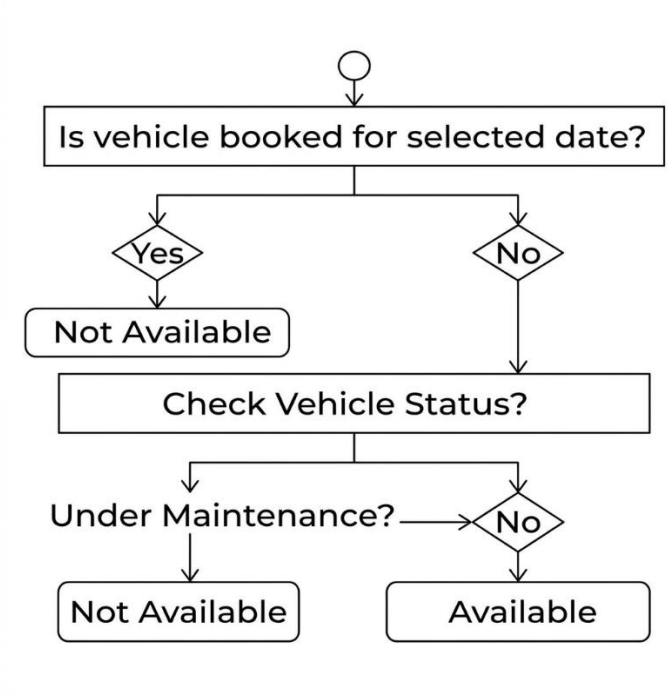



Table 3.6 ***Vehicle Availability Decision Tree***

# CHAPTER THREE: SYSTEM IMPLEMENTATION (DEVELOPMENT, TESTING AND DEPLOYMENT)

## 3.1 Introduction

chapter provides a comprehensive account of the system's transition from conceptual design to a fully functional Car Rental Management System, outlining the entire development process from the setup of the Integrated Development Environment (IDE) and configuration of the technology stack to the implementation phase. It discusses the iterative development of the front-end, including role-based interfaces for staff and clients that support activities such as vehicle booking and customer management, as well as the back-end logic that enables real-time vehicle availability tracking, rental processing, and revenue monitoring. Additionally, the chapter describes the database implementation for managing system data, the testing procedures carried out to ensure system reliability, performance, and data integrity, and finally the deployment strategy used to make the system accessible and operational for end-users.

## 3.2User Interface Development

This section presents the development of the **Car Rental Management System**, which is divided into two main components: the user interface (user side) and the system logic (back-end). The user interface enables different users, such as staff and clients, to interact with the system through various pages including login, registration, dashboard, booking, and report pages. The logic side handles the processing of user requests, data management, and system operations such as vehicle availability, rental processing, and revenue tracking.

### 3.2.1 Login Page Development

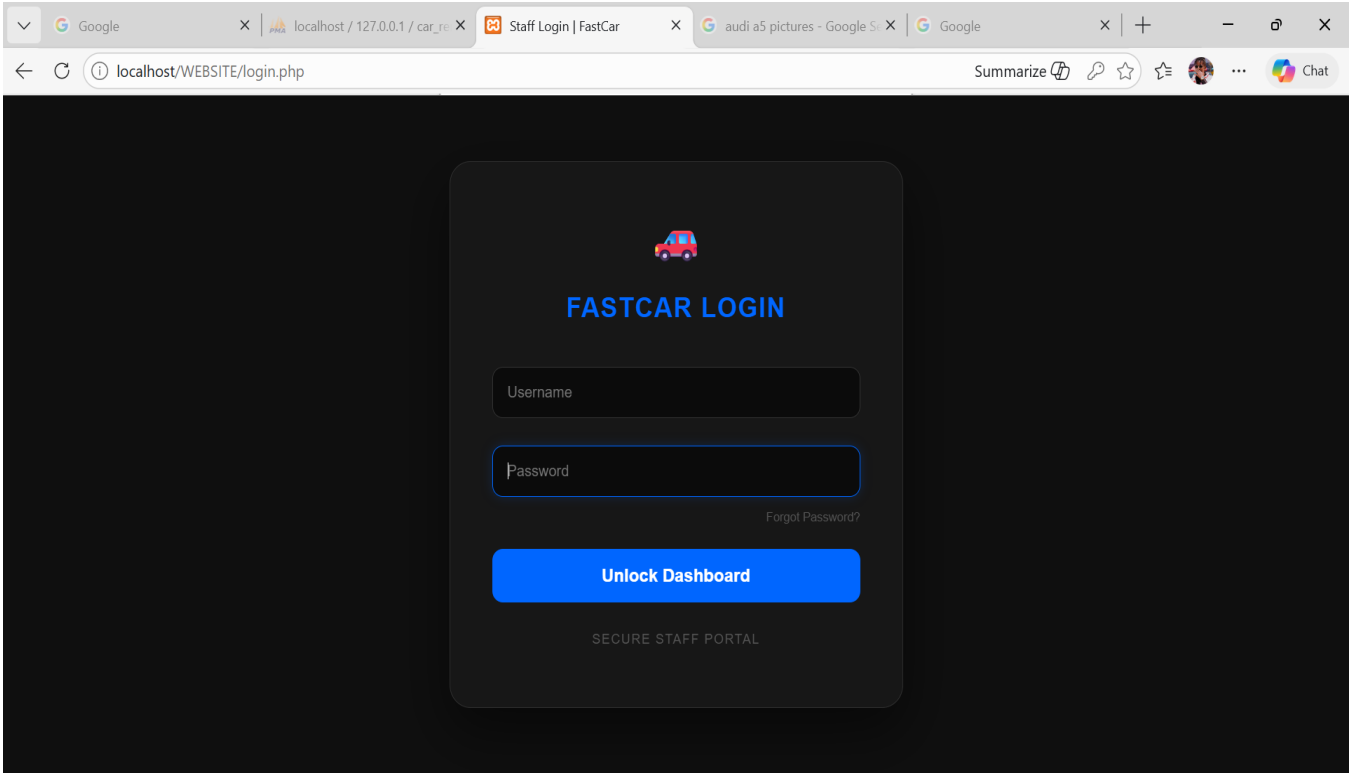



*Figure 1.1 Login Page Development*

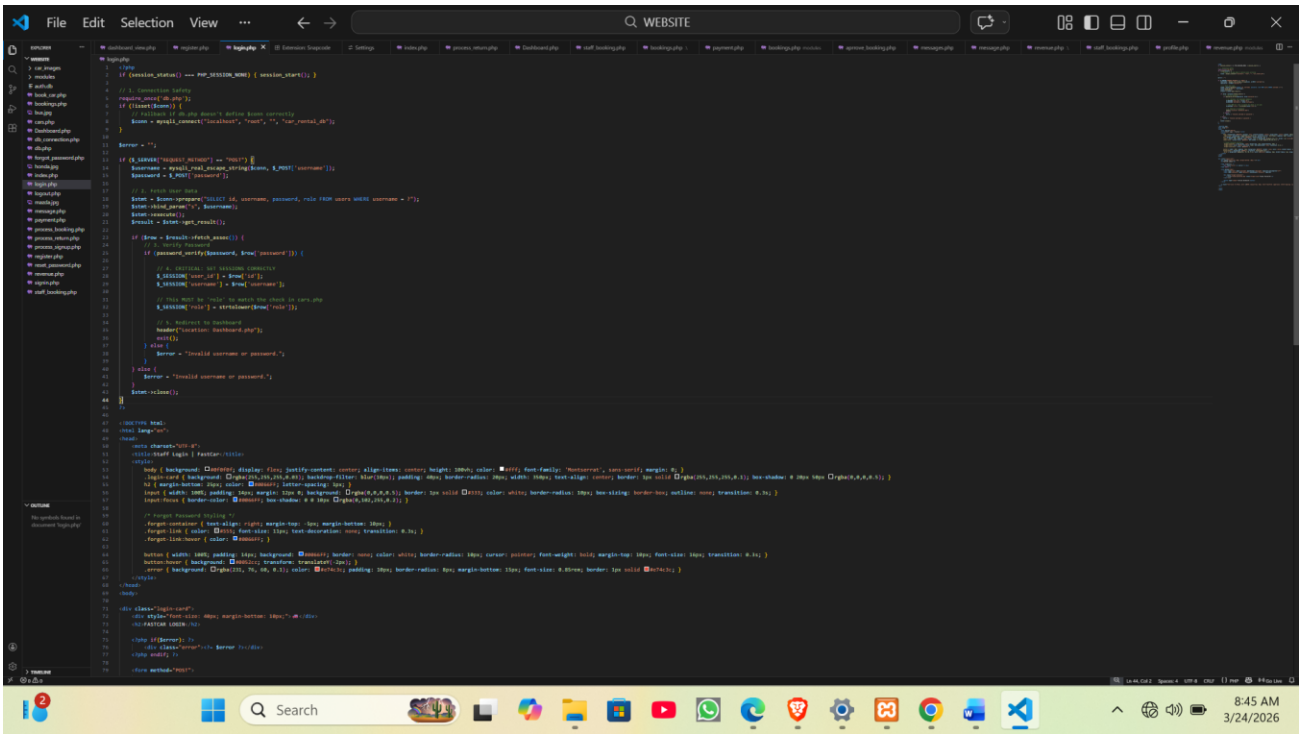


*Figure 1.2Login Page code*

3.2.2 Registration / Sign-Up Page

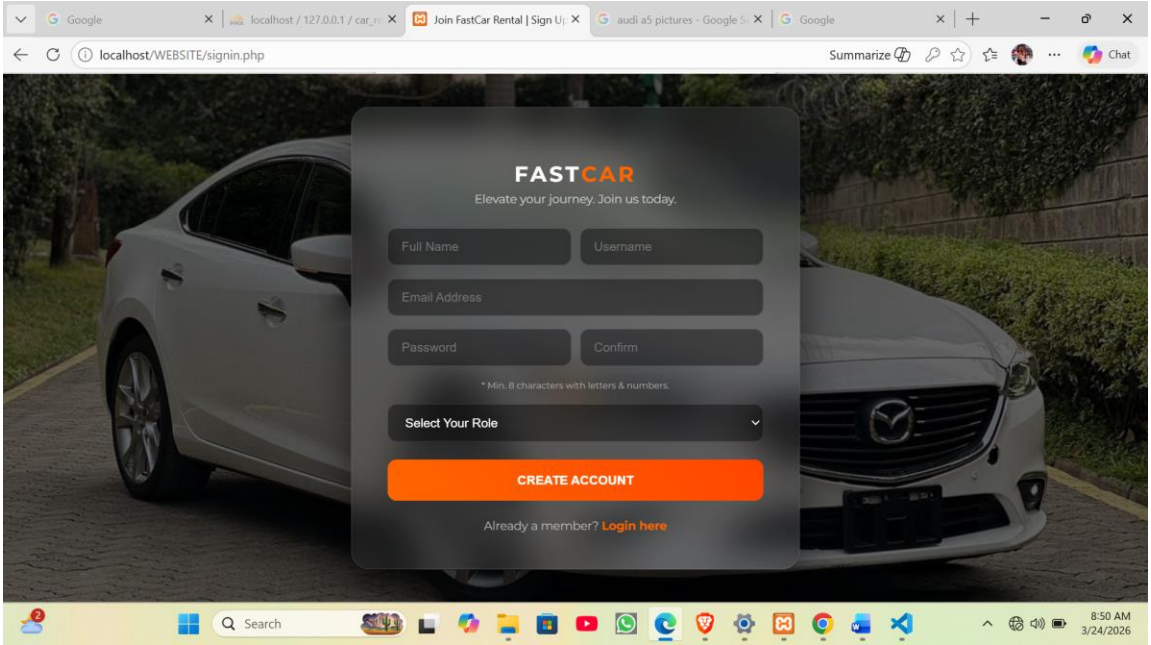



*Figure 1.3 Sign-Up Page*

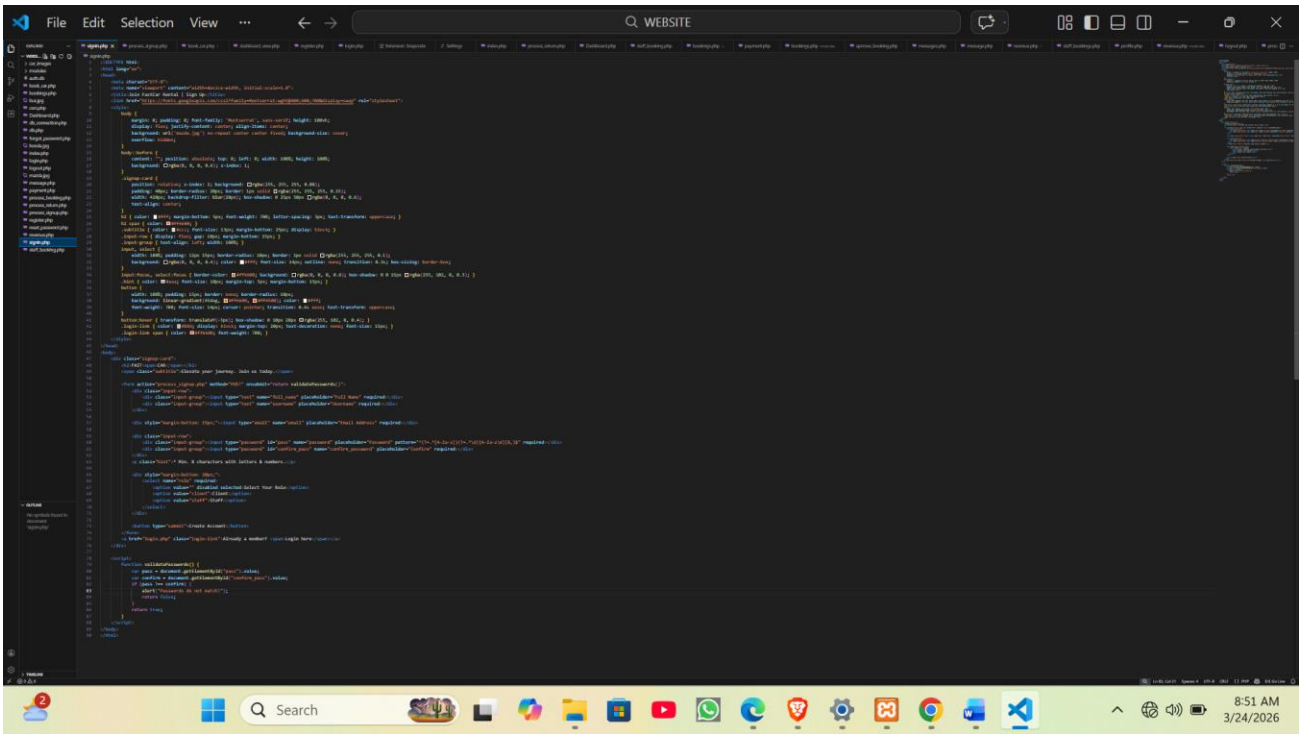



*Figure 1.4 Sign-Up code*

3.2.3 Dashboard Page

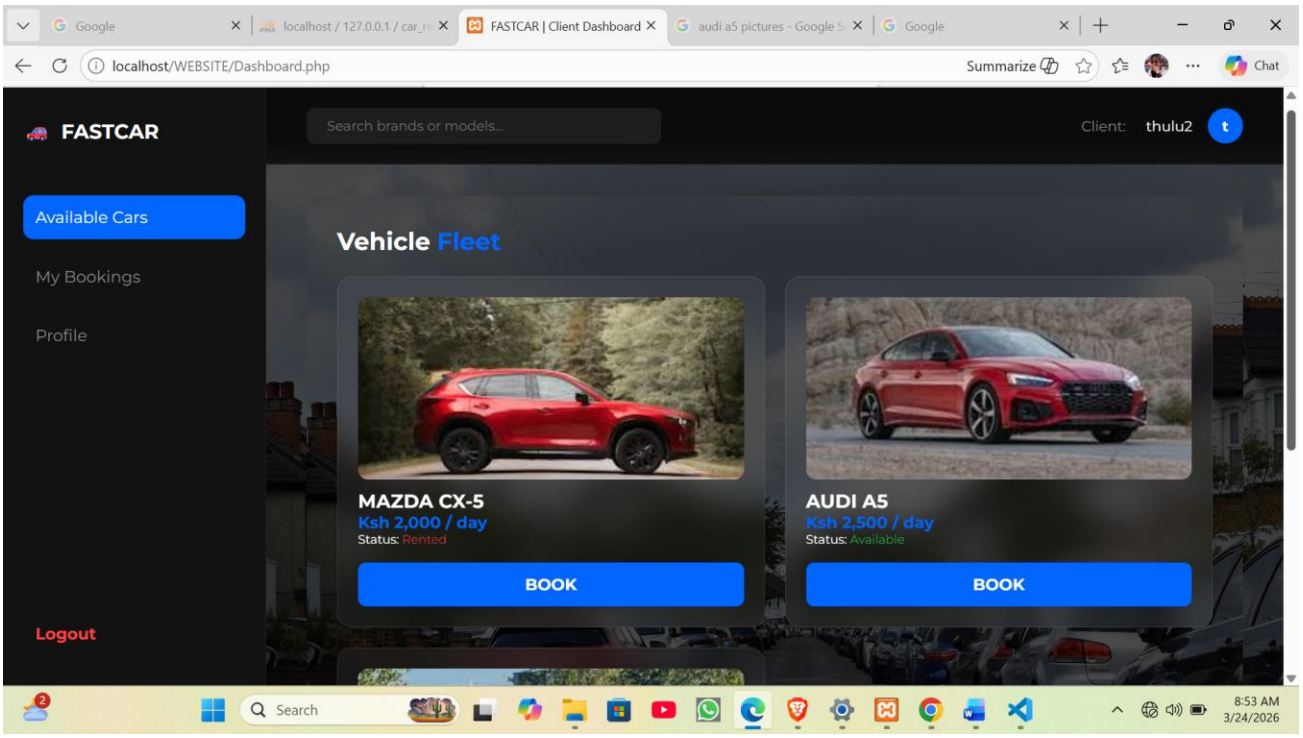



*Figure 1.5dash board page*

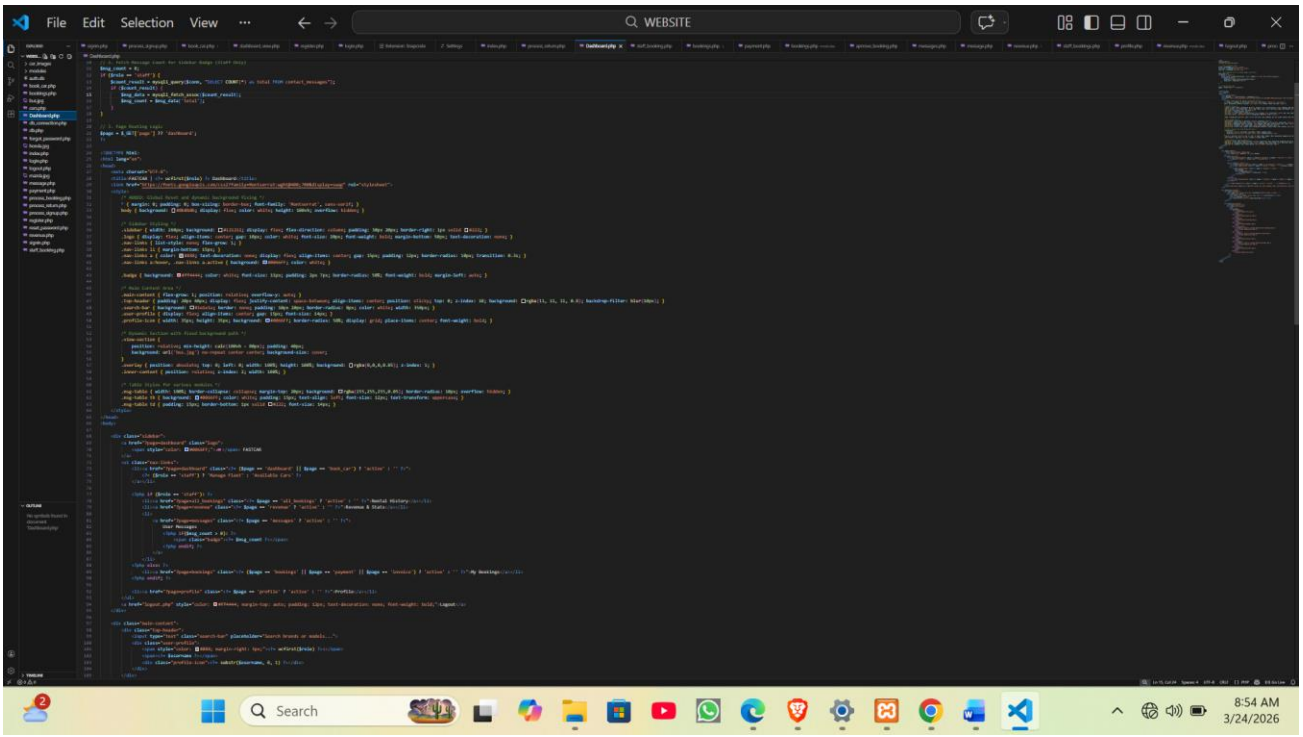



*Figure 1.6 dashboard code*

### 3.2.4 Vehicle Management Page

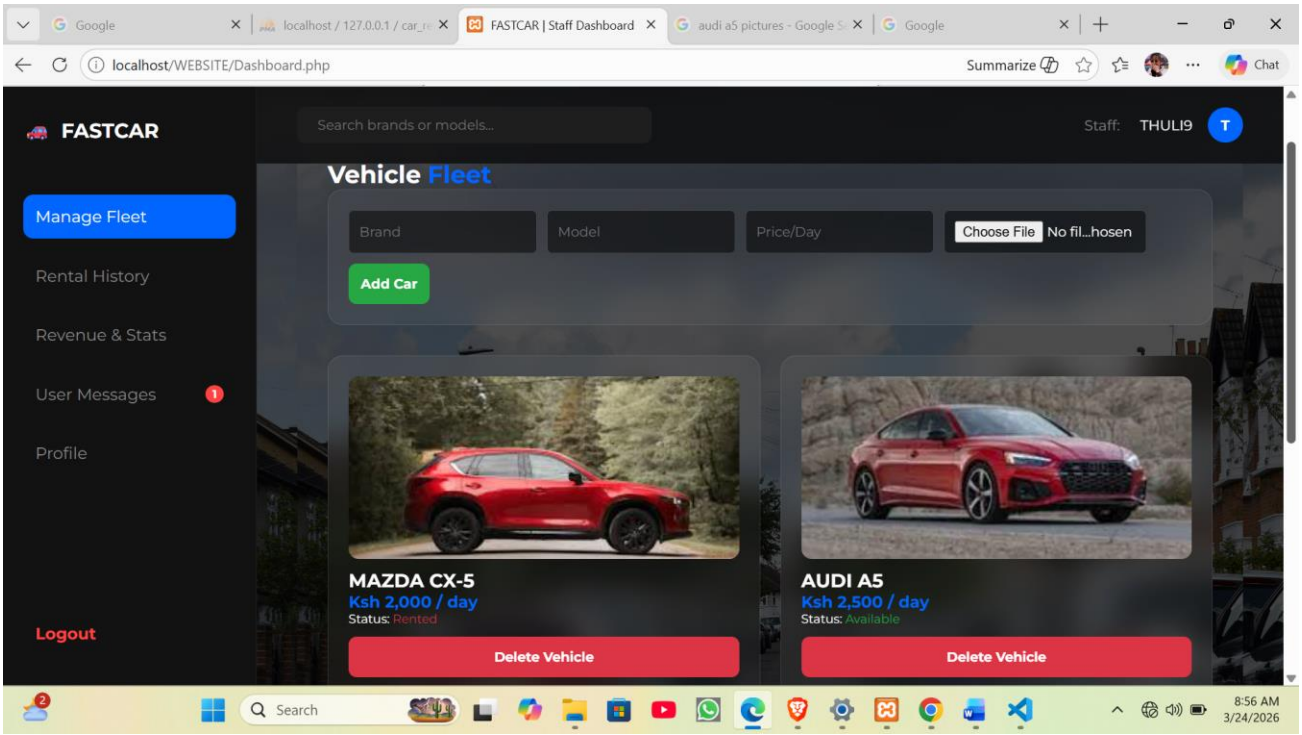



*Figure 1.7 Vehicle Management Page*

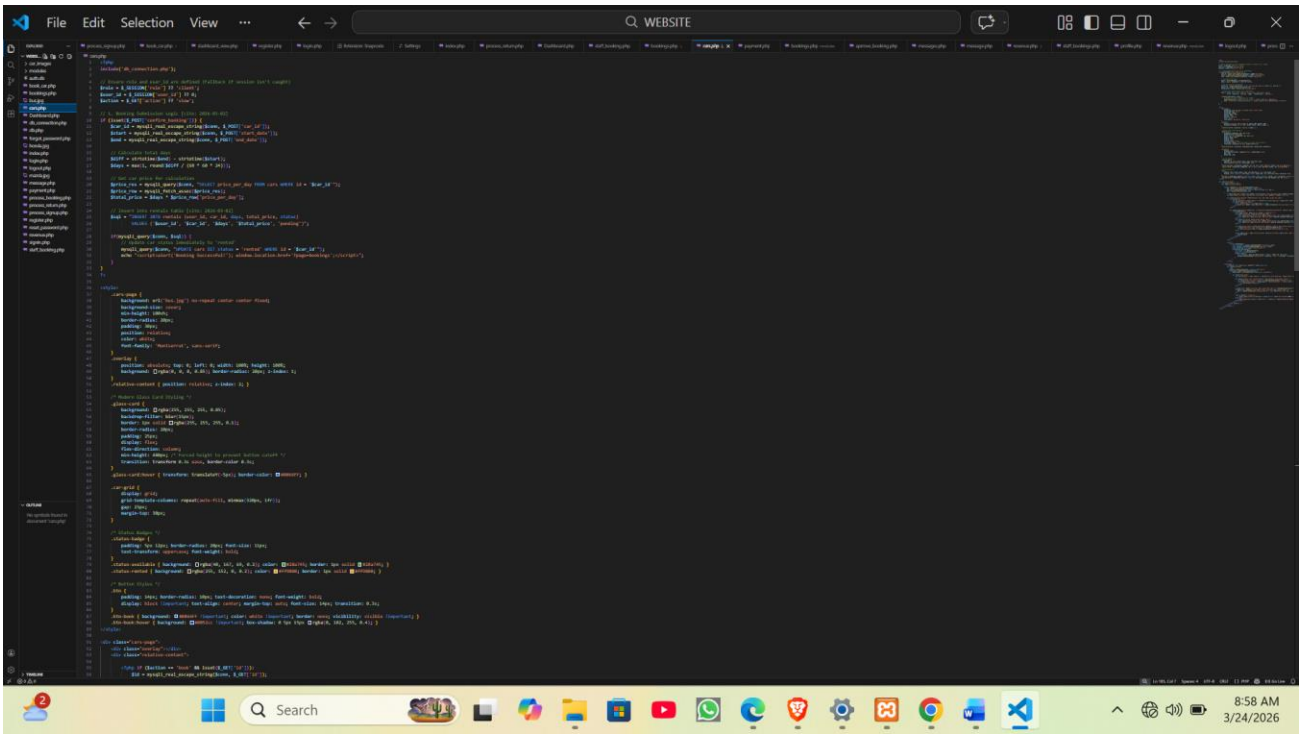



*Figure 1.8 Vehicle Management Page*

3.2.5 Booking / Rental Page

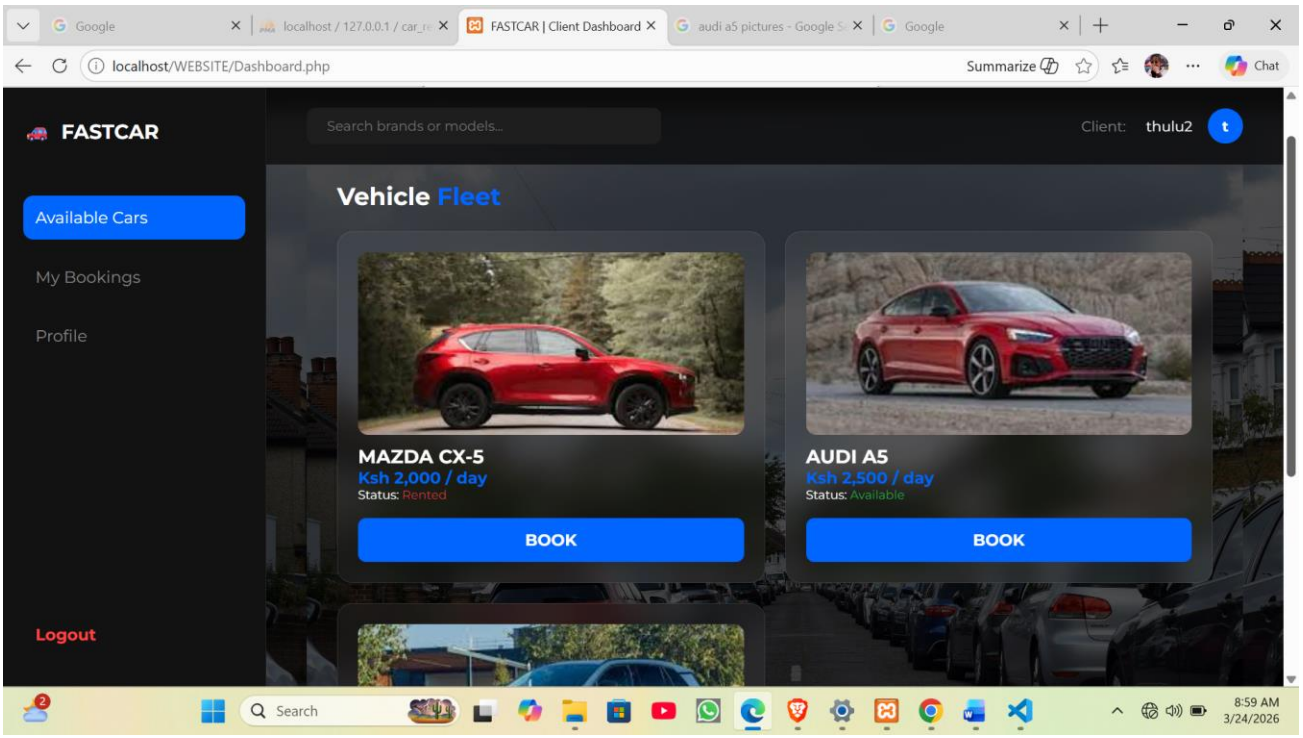



*Figure 1.9 Rental Page*

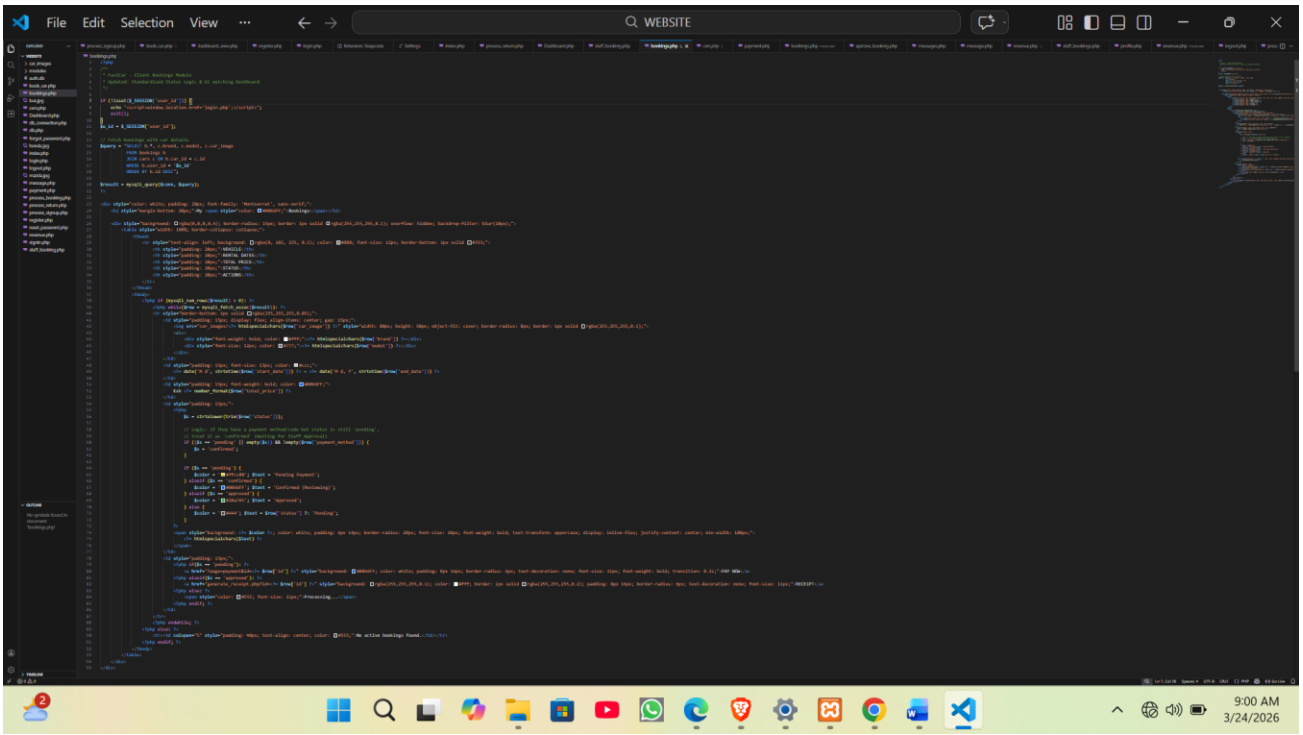



*Figure 2.*9 Rental Page

### 3.2.6 Revenue Page

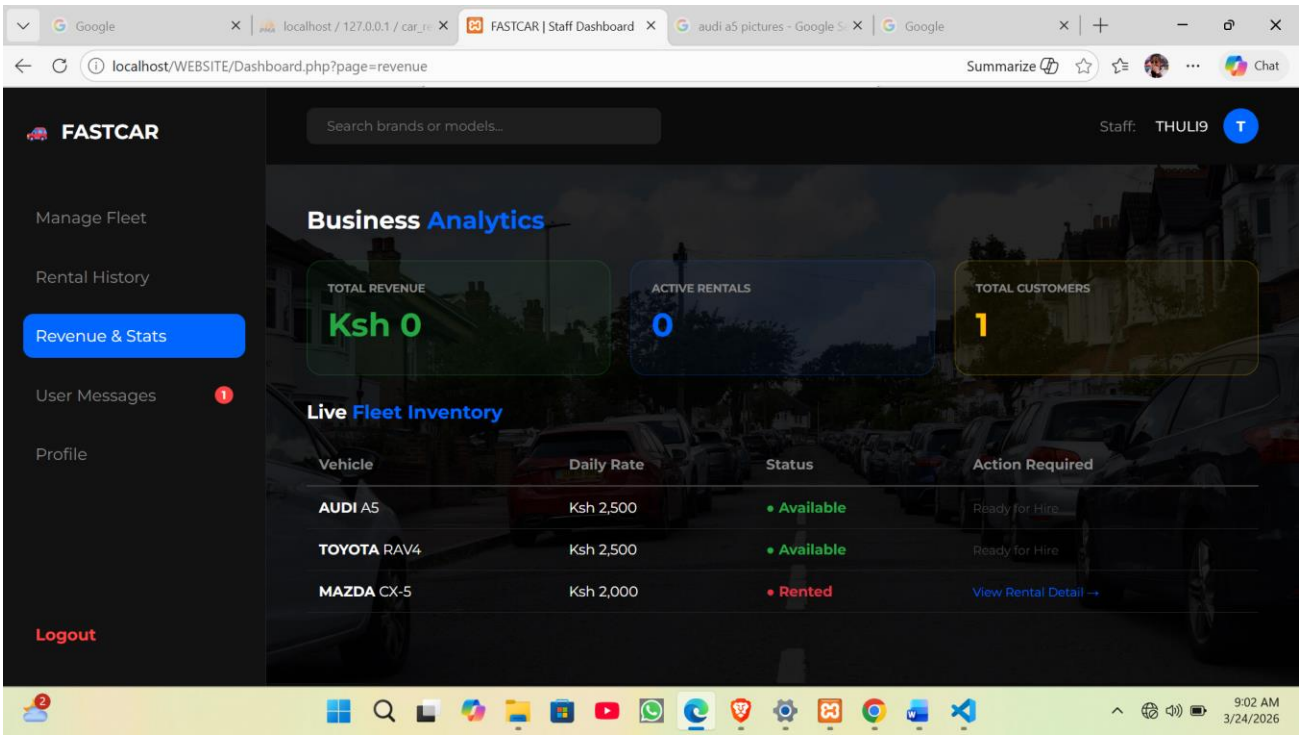



*Figure 2.1revenue page*

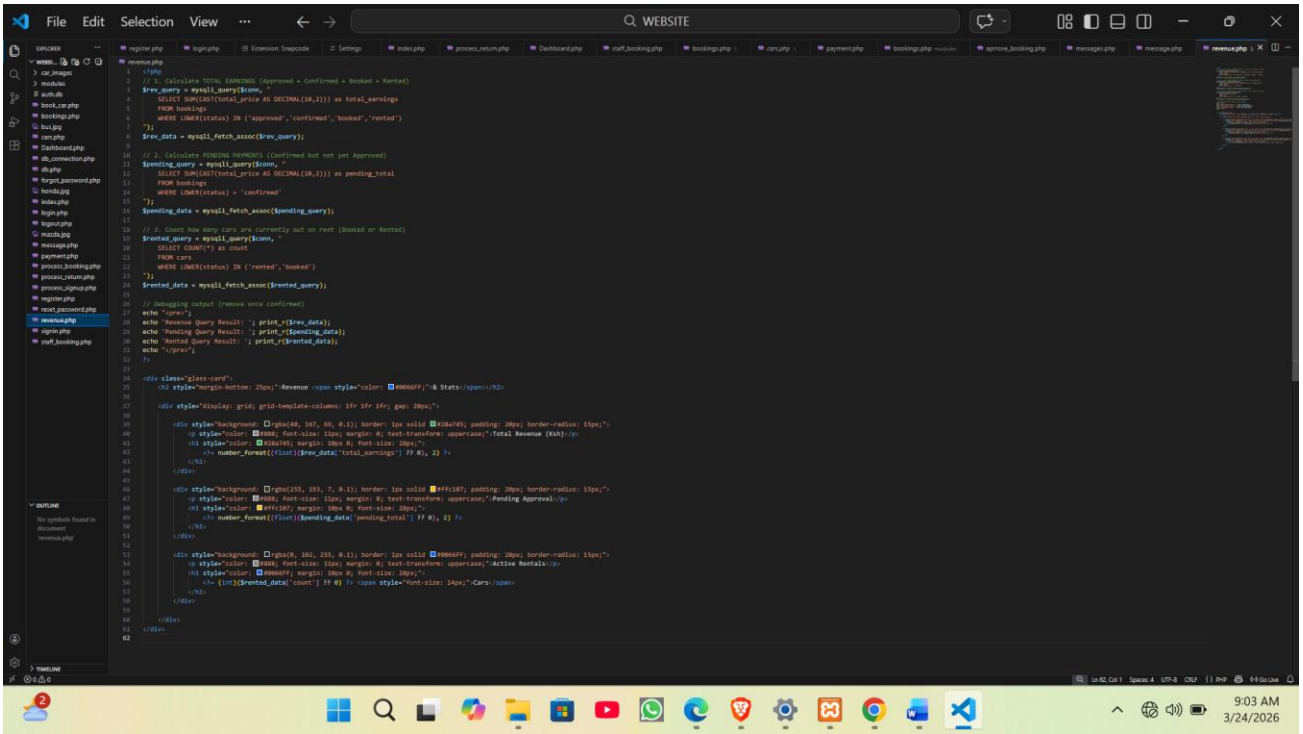



*Figure 2.21revenue page*

### 3.2.7 Database

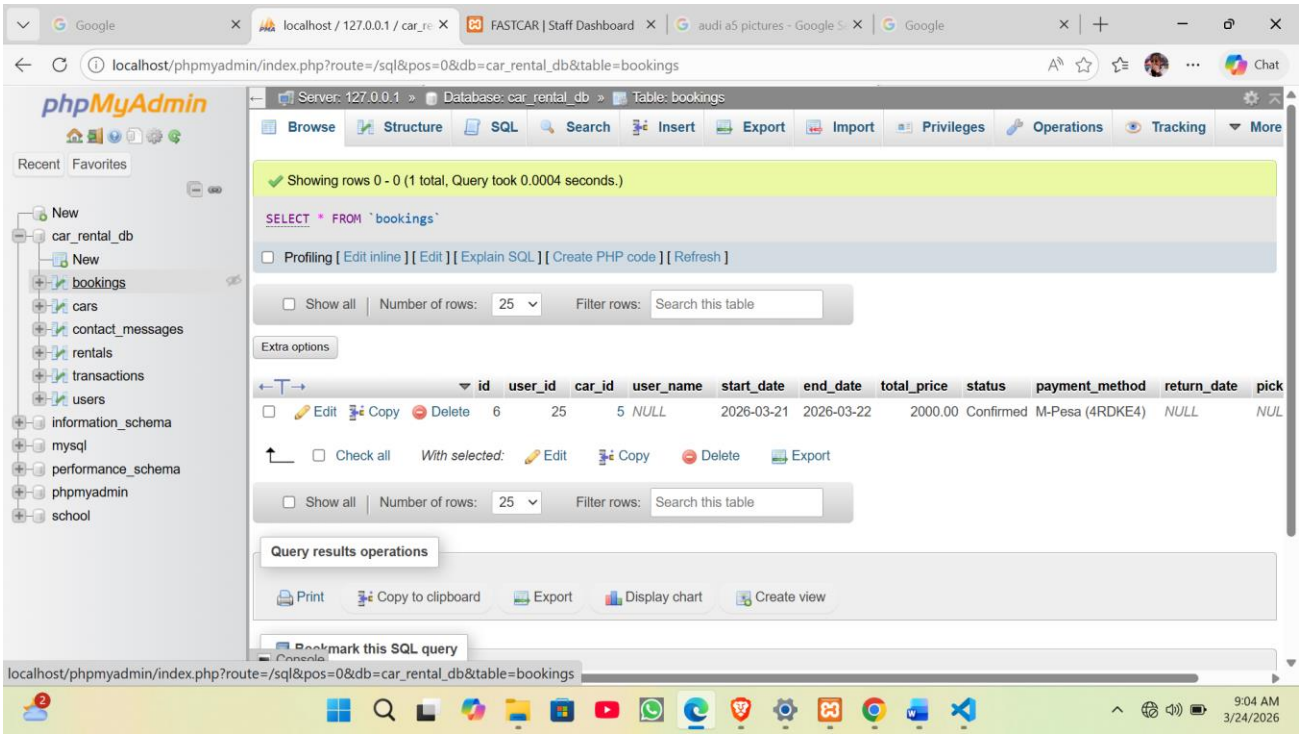



*Figure 2.3 database*

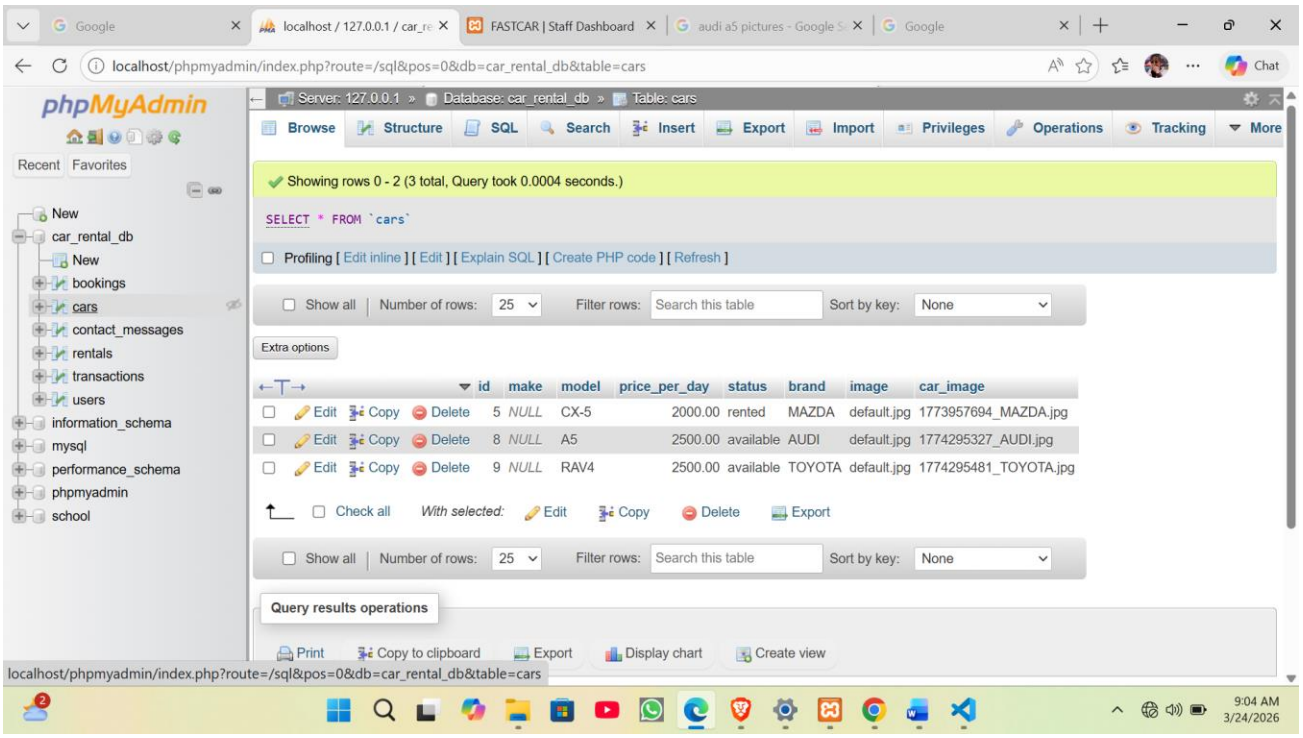



*Figure 2.4 database*

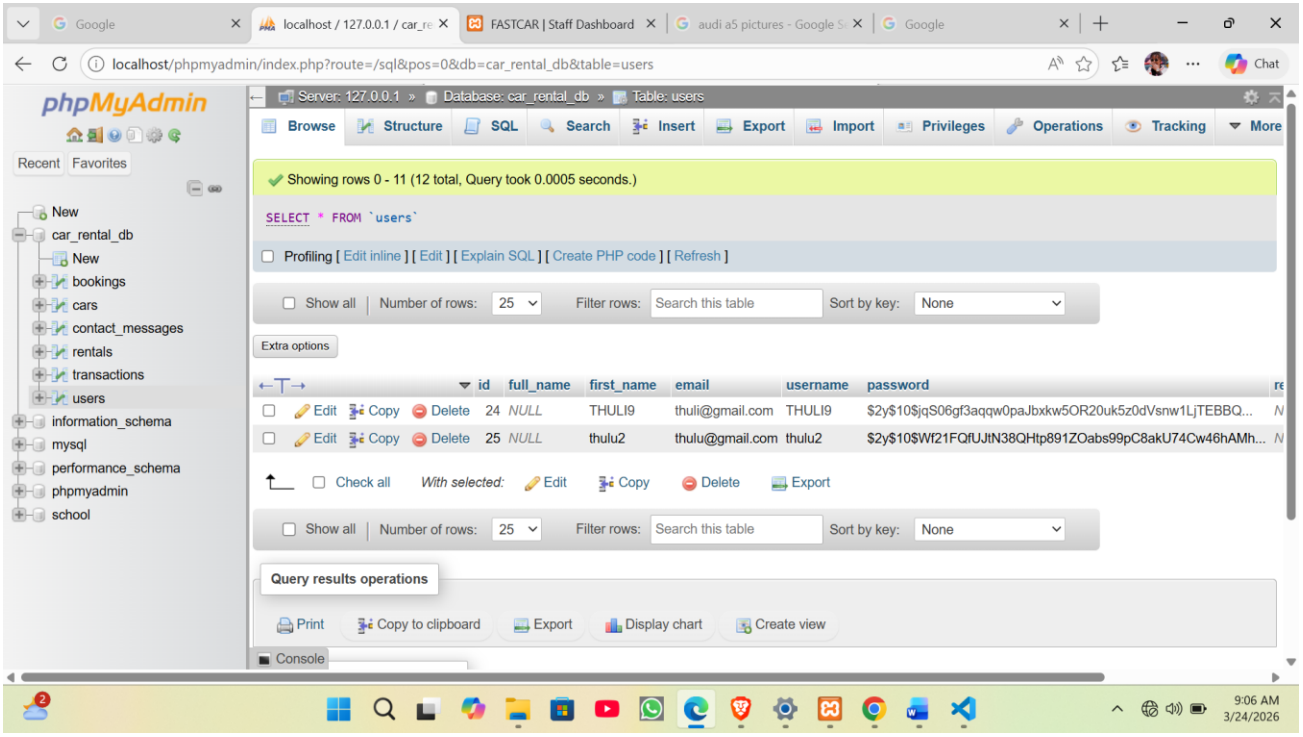



*Figure 2.5 databased*

3.3 Logic Development

3.3.2 signup code

Allows a new user to create their new account and select their role.

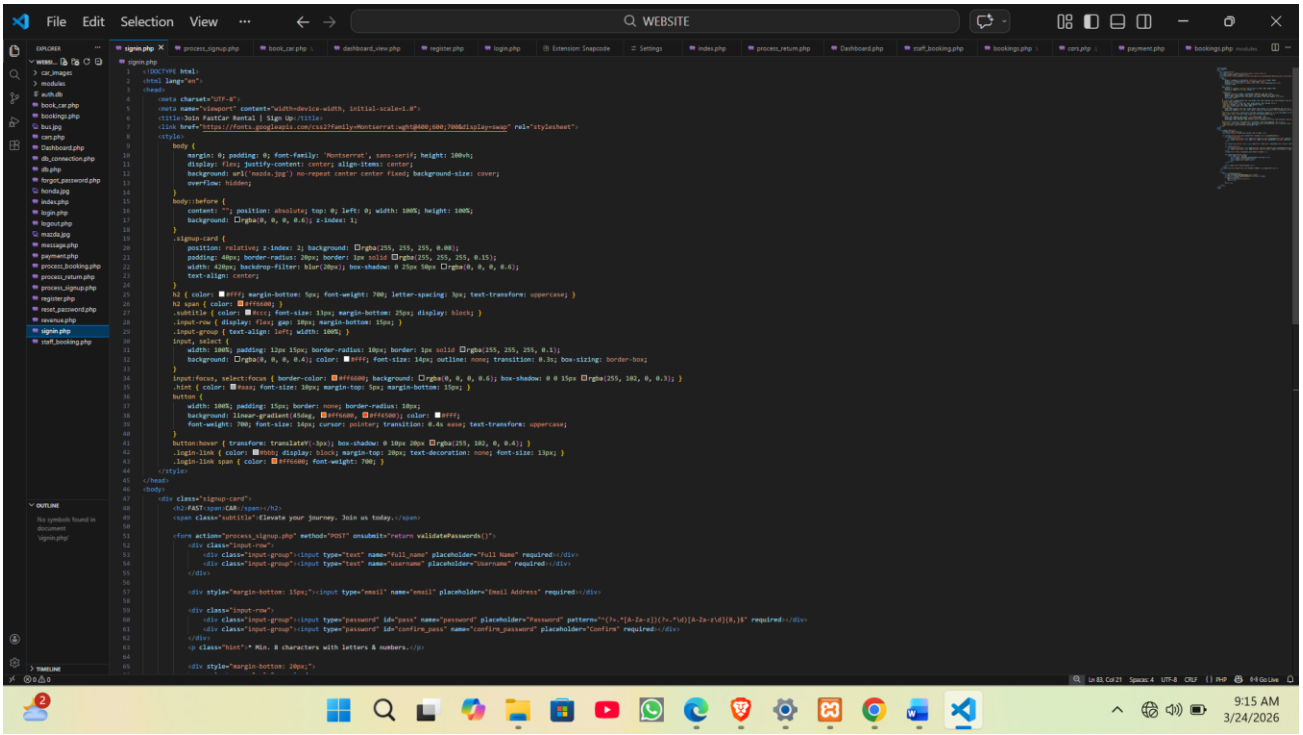



*Figure 3.2 signup code*

3.3.3 login page.

It allows the staff or users to log in the dashboard according to their roles.

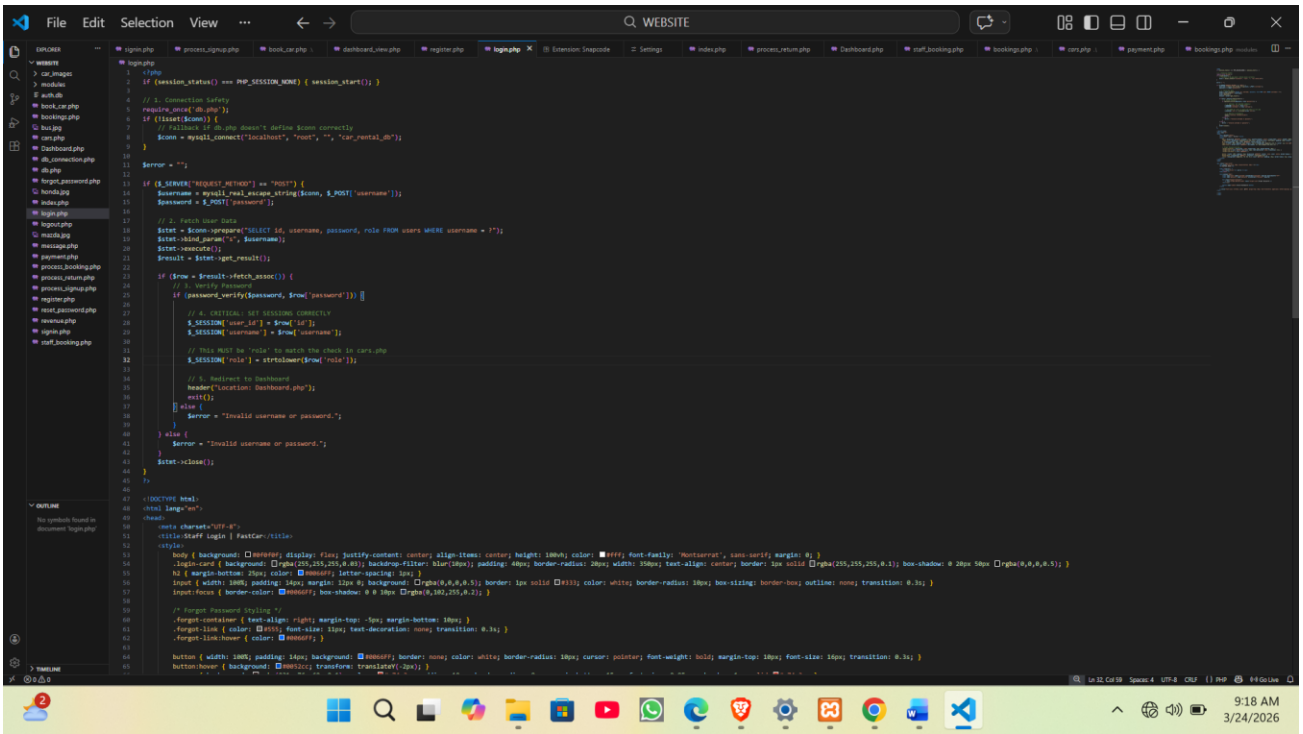



*Figure 3.3 login code*

### 3.3.4 dashboard code

The client are able to see available cars book and make payments, create their profile and see their booking history.

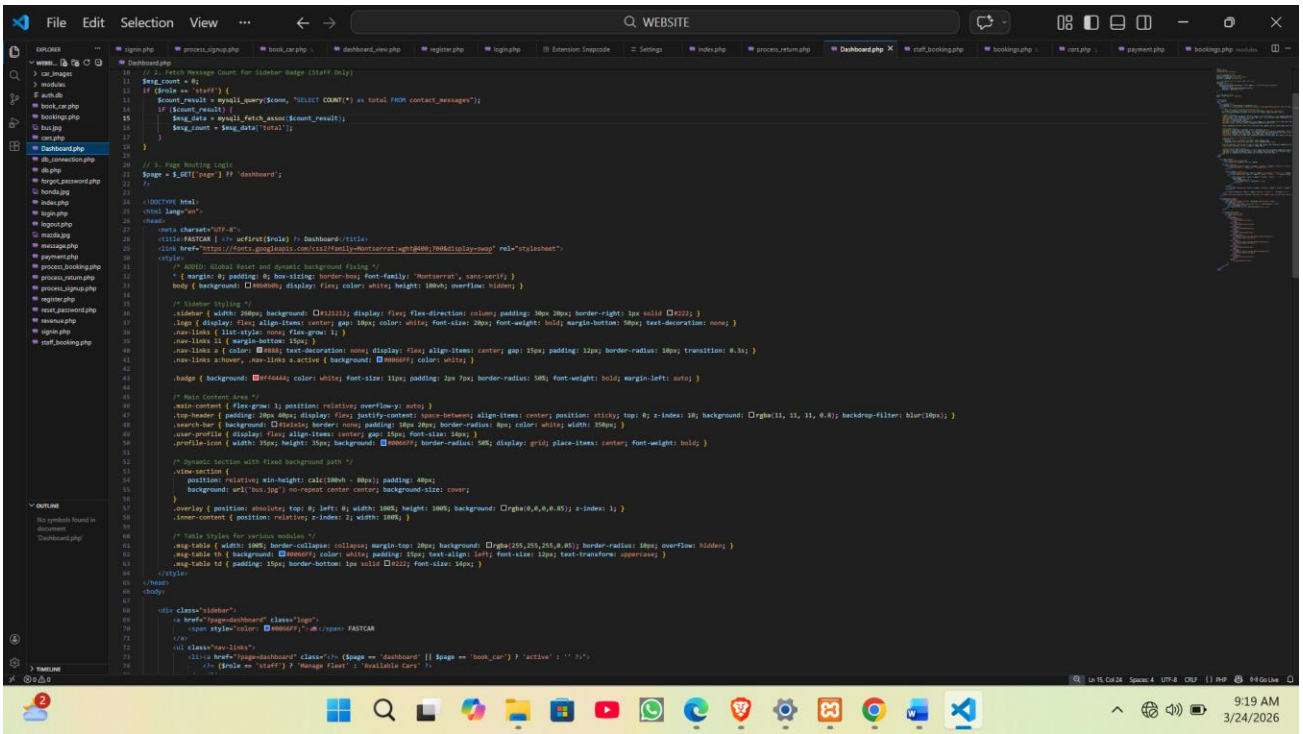


*Figure 3.4 dashboard code*

### 3.4 Deployment

The system is deployed on **InfinityFree**, a popular web hosting service that provides the necessary infrastructure for PHP-based applications. It offers a LAMP (Linux, Apache, MySQL, PHP) stack, which is compatible with the environment used during the development phase in XAMPP.

Live Project URL**:**

https://dash.infinityfree.com/accounts/if0_41463335/domains/carrentals.infinityfreeapp.com

## *CHAPTER FOUR: CONCLUSION AND RECOMMENDATION*

### Conclusion

The car rental management system project was developed to solve challenges associated with manual operations such as poor record keeping, data redundancy, and delays in processing bookings. Chapter One introduced the background, problem statement, objectives, and significance of the project, clearly outlining the need for an automated system to improve efficiency and accuracy. Chapter Two reviewed existing systems and relevant technologies, highlighting their strengths and weaknesses while identifying gaps that the proposed system aimed to address, such as better data management and user-friendly interfaces.

Chapter Three focused on the system design and methodology, including tools, techniques, and system requirements used during development. It explained how diagrams like flowcharts and data models guided the structure of the system. Throughout the project, new skills such as database design, system analysis, and programming were learned, while challenges like debugging errors, designing an effective system structure, and managing time were encountered. Despite these difficulties, the project provided valuable experience and improved both technical and problem-solving abilities.

### Recommendation

If you decide to further improve this car rental management system, one major area to focus on is the payment

module, which was not fully implemented in the current version due to time and resource limitations. The system is fully functional in terms of booking, customer management, and record keeping; however, payments are handled manually outside the system. You can enhance this by integrating modern payment solutions such as M-Pesa, bank card payments, or online payment gateways to allow customers to make secure and convenient transactions directly within the system. This would not only improve efficiency but also provide real-time payment tracking and reduce errors associated with manual handling.

In addition, you can consider improving other aspects of the system such as implementing barcode or QR code scanning for faster customer and vehicle identification, as well as developing a web or mobile-based platform to allow remote access and booking. Strengthening system security through advanced authentication methods and adding reporting and analytics features would also be valuable upgrades. These improvements would build upon the existing working system and transform it into a more modern, efficient, and scalable solution.

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